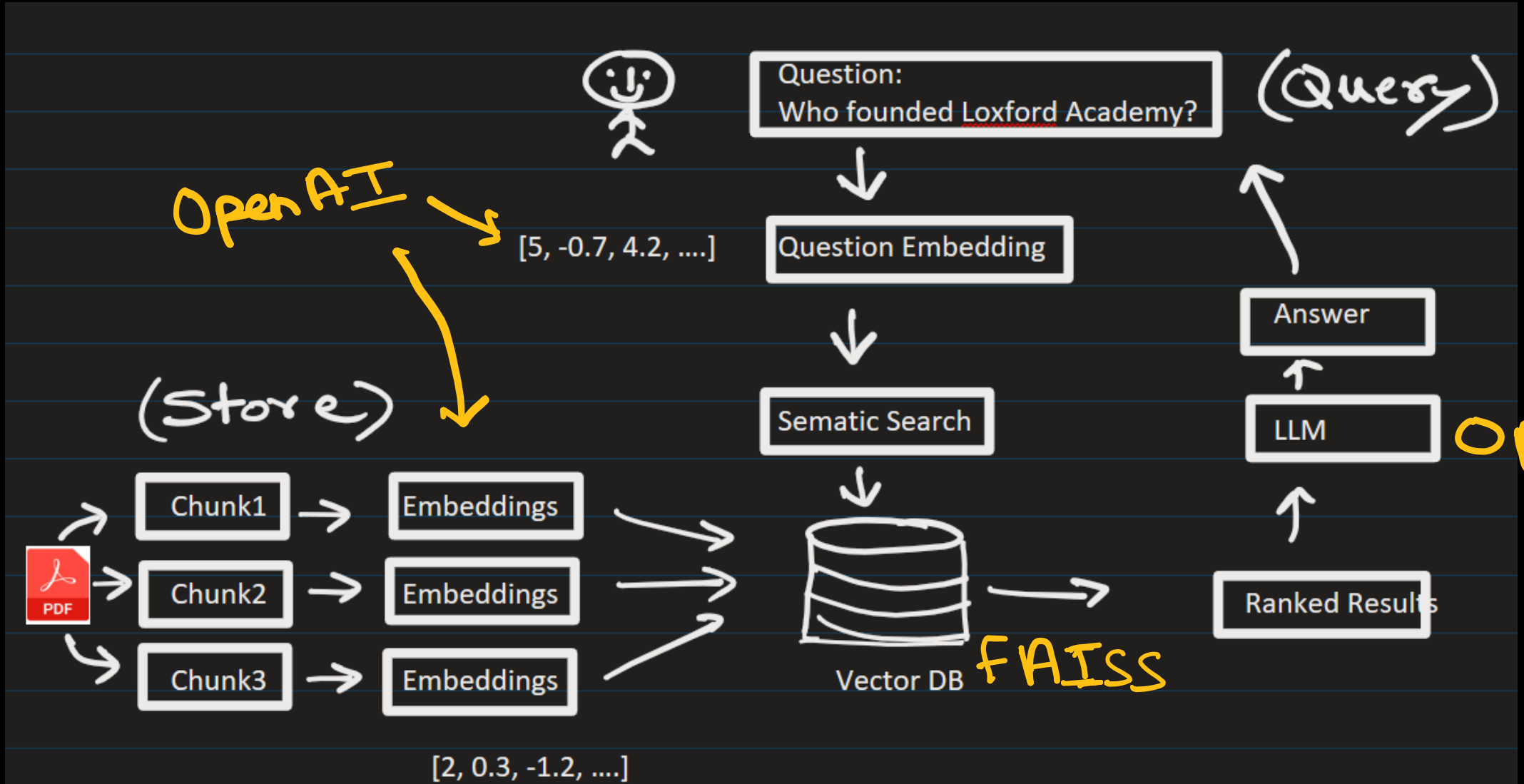
# RAG (Retrieval-Augmented Generation)

RAG (Retrieval-Augmented Generation):
Phase 1) Populate the vector DB
Phase 2) User asks questions.



Steps:

1) Get an OpenAI API key from their website

2) I am using python3.12. Install following packages:

> python –m pip install streamlit openai faiss-cpu pypdf numpy python-dotenv

It installed following version

- streamlit==1.54.0
- openai==2.29.0
- faiss-cpu==1.13.2
- pypdf==6.9.1
- numpy== 2.4.3

You can manually install above version:

> python –m pip install streamlit==1.54.0 openai==2.29.0 faiss-cpu==1.13.2 pypdf==6.9.1  numpy== 2.4.3 python-dotenv

**3) Go to directory that contains app.py and run the program:**
**> python -m streamlit run .\app.py**

**4) Goto browser and load a PDF document-Loxford-company:**

**Ask question related with the PDF doc:**
Q1) Who is the current CTO of Loxford Company ?
Answers correctly

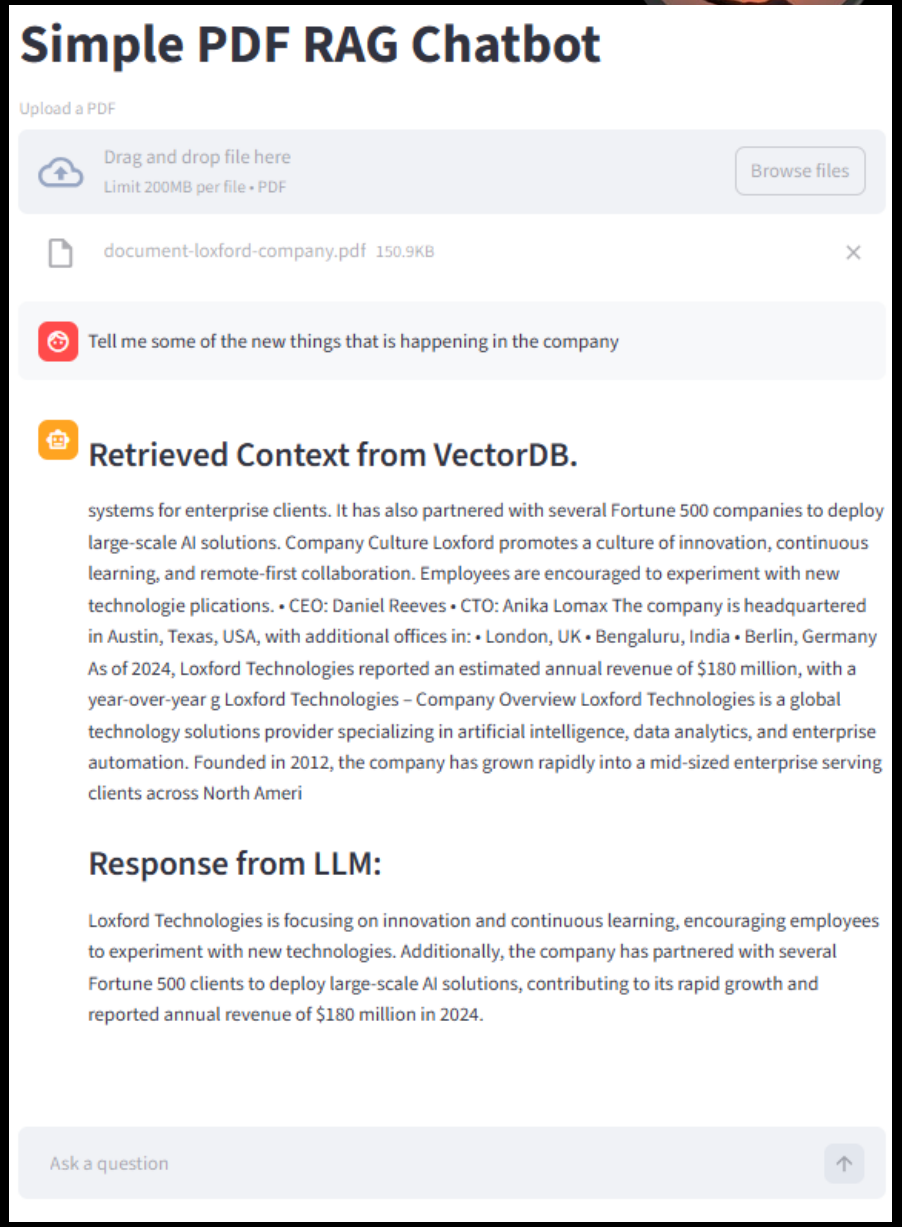
Q2) Tell me the range  of industries it serves
Answers correctly

Q3) How many professional it employs ?
Change wording -> How many people are currently working here?
Answers correctly

Q4) Tell me some of the new things that is happening in the company
Answer is not perfect. Missing the context somewhat.

There is some room for improvement here:

- Increase the chunks size
- Use overlapping in the chunks
- Use semantic splitting
- Use paragraph/header splitting, etc.



# EXTRA

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Fl;ds